1. Suppose that a is declared as `int a[10]`. Give the contents of the array after the following two statements are executed in sequence.

```
for ( i = 0 ; i <= 9; i++ )
    a[i] = 9 - i;
for ( i = 0 ; i <= 9; i++ )
    a[i] = a[a[i]];
```

2. Write a program to fill in a two-dimensional array of 0-1 values by setting `a[i][j]` to 1 if `i == j`, and to 0 otherwise.

3. Declare a one-dimensional array, `name`, that contains 20 characters. If each character occupies one "cell" in memory, and the base address of `name` is 1000, what is the address of the cell referenced in the following statement? `name[9] = 'A';`

4. Declare a one-dimensional array type, `WeatherTypeList`, of `WeatherType` elements, that has 12 elements (representing the months). Each element should have information about average high temperature and average low temperature. How can we assign the value 30 to the avg high temp of August? Also, declare a two-dimensional array, `DecadeWeather`, of `WeatherType` elements. Assign the value 26 to the avg low temp member of the March 2006 entry.

5. Construct a suitable data type that is appropriate for each of the following:

   (a) Chessboard

   (b) Point in a plane

   (c) A list of famous quotations

   (d) An electronic address book (name, address, phone)

   (e) A collection of hourly temperatures for a 24-hour period

6. The following code has three separate errors, each of which could cause an infinite loop. Can you identify them?

```
void increment(int i){
    i++;
}
int main(){
    int count = 1;
    while (count < 10){
        cout<< count;
        increment(count);
        cout<< count;
    }
}
```

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